# Sahaj Bhadja

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## Summary

Quantitative Engineer with 2+ years of experience building **Python data pipelines**, **distributed analytics and classification models**, reusable tests, scalable data applications, and production-quality model evaluation workflows.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Data Retrieval, Operating Systems, Software Engineering

## Experience

**USC Dornsife** | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**

- Achieved 99%+ reliability across 5K+ monthly jobs through **incremental data pipelines**, implementing validation, reconciliation, error handling, cloud storage, CI/CD releases, and stable production observability reliably at scale.
- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant analytics platform supporting 500+ users reliably.

**Grad8** | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**

- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Maintained 90%+ scoring consistency across 10K+ submissions through **structured model evaluation**, model switching, fallback paths, persistent results, and reviewer controls supporting reproducible testing and reliable analytical workflows.

**USC AD Labs** | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**

- Improved long-horizon trajectory planning by 18% in simulation by building a **quantitative modeling pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and reproducible evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through a **spatial-temporal model** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex simulation scenarios.

**Aasma Technology Solutions** **August 2023 – July 2024**
*Software Developer Engineer - (SecureSync)* | *India*

- Increased pipeline efficiency by 68% through **AWS Glue and S3 ETL**, transforming JSON, CSV, and Parquet inputs into normalized downstream datasets while reducing storage costs by 15% reliably.
- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, generating failure reports, coverage summaries, and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows.

**Cyber Security and Data Mining: Detecting Malicious URL** **May 2023 – August 2023**
*Data Analyst - Research Fellowship* | *India*

- Achieved 96%+ classification accuracy across 650K+ URLs by comparing **Random Forest, CNN, RNN, LSTM, Bi-LSTM, and GRU models** within a reproducible machine-learning evaluation pipeline for phishing and malware detection.
- Engineered 21 behavioral indicators covering URL length, IP presence, and suspicious keywords, translating noisy inputs into **classification features** that exposed phishing and malware patterns reliably at scale.

## Skills

**Languages:** Python, SQL, Java, C++, JavaScript, TypeScript
**Big Data & Processing:** Spark, Databricks, Hadoop, AWS Glue, AWS S3, ETL, Distributed Computing
**Modeling & Analytics:** Classification, Random Forest, CNN, RNN, LSTM, Feature Engineering, Model Evaluation, Statistics
**Data & Applications:** PostgreSQL, MySQL, MongoDB, NoSQL, AWS RDS, React, Data Modeling, Query Optimization
**Quality:** Git, Jenkins, Docker, CI/CD, Unit Tests, Integration Tests, Test Automation, PyTest, SDLC, Documentation, Observability

## Projects

**Enterprise Data Quality Framework - Design Concept**

- Designed a **distributed data-quality framework** concept with reusable schema, completeness, reconciliation, lineage, and control checks across partitions, plus exceptions, regression tests, audit evidence, and configurable risk thresholds.

**Portfolio Risk Model Testing Platform - Design Concept**

- Designed a **Python risk-testing platform** concept with classification baselines, time-based validation, drift checks, scenarios, reproducible data, approvals, documentation, and React dashboards for clear analytical review by stakeholders.

**FinNexus** | Live

- Built a **secure fintech platform** supporting 50K+ users and 1M+ transactions with Plaid bank linking, token-based access controls, real-time synchronization, advanced filtering, Kubernetes deployment, and MongoDB-backed data access.